SPECTRUM

Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models

1 Forcing **2 Component models** **3 Sum and test** **4 Forecast to 2100**

Global mean surface temperature

GMST

observed to 2025
AR6 paths: 2, 3, 4 °C

Greenland ocean T
via Arctic amplification

West Antarctica *storyline*
slow/fast storyline, independent of emissions

East Antarctica *adopted*
snowfall sensitivity from literature

Antarctic Peninsula *calibrated*
linear in warming · IMBIE-3

Greenland SMB *emulated*
MAR emulator, 7 CMIP6 models

Greenland discharge *calibrated*
ocean-forced, lagged · Mouginot 2019

Glaciers *calibrated*
linear in warming · GlaMBIE

Terrestrial water storage *adopted*
IPCC AR6

Thermosteric *calibrated*
two-layer ocean · NOAA 0–2000 m

2000 members under 2, 3, 4 °C

observed GMSL *withheld*

West Antarctica *tail, all scenarios*

4 °C
3 °C
2 °C

GMSL

1950 2025 2100

schematic stack and fan

Test against observed GMSL, withheld from every calibration

- ✓ closes the altimetry budget, 1993–2025
- ✓ hindcasts GMSL back to 1950

Two dimensions of risk

Emissions set the warming-driven components

West Antarctica sets the tail

Lane style: solid = calibrated by Bayesian inference against independent multi-decade records · hatched = emulated · dashed = adopted